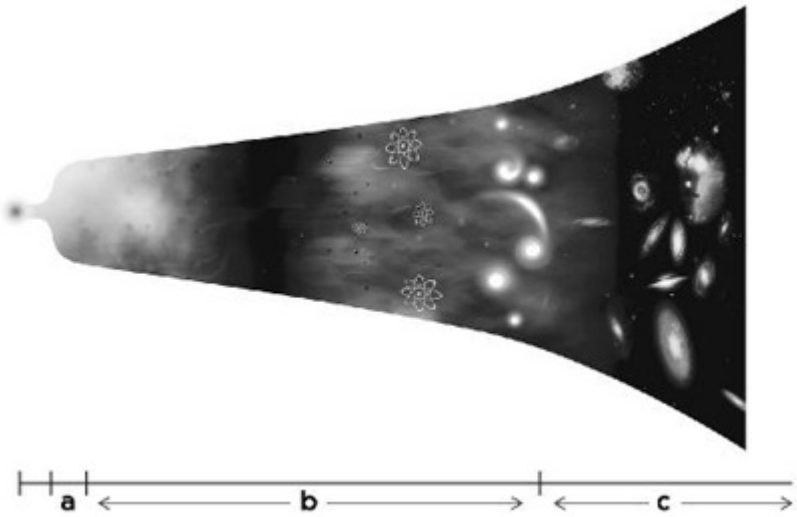


Figure 10.6 A time line of cosmic evolution. **(**a**)** Inflationary burst. **(**b**)** Standard Big Bang evolution. **(**c**)** Era of accelerated expansion.

Puzzles and Progress

With these discoveries, it thus seemed manifest that the pieces of the cosmological puzzle were falling into place. Questions left unanswered by the standard big bang theory—What ignited the outward swelling of space? Why is the temperature of the microwave background radiation so uniform? Why does space seem to have a flat shape?—were addressed by the inflationary theory. Even so, thorny issues regarding fundamental origins have continued to mount: Was there an era before the inflationary burst, and if so, what was it like? What introduced an inflaton field displaced from its lowest-energy configuration to initiate the inflationary expansion? And, the newest question of all, why is the universe apparently composed of such a mishmash of ingredients—5 percent familiar matter, 25 percent dark matter, 70 percent dark energy? Despite the immensely pleasing fact that this cosmic recipe agrees with inflation’s prediction that the universe should have 100 percent of the critical density, and although it simultaneously explains the accelerated expansion found by supernova studies, many physicists view the hodgepodge composition as distinctly unattractive. Why, many have asked, has the universe’s composition turned out to be so complicated? Why are there a handful of disparate ingredients in such seemingly random abundances? Is there some sensible underlying plan that theoretical studies have yet to reveal?